



Multifrequency signal analysis by Interpolated DFT method with maximum sidelobe decay windows

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ABSTRACT

In this paper, the estimation of a multifrequency signal parameters by the Interpolated Discrete Fourier Transform (IpDFT) method with maximum sidelobe decay windows is investigated. First, the theoretical expressions of ones of the most important parameters of a H -term maximum sidelobe decay window ($H \geq 2$) are derived. In the following, the analytical formulas for estimating the parameters of a multifrequency signal are derived. The influence of a white Gaussian noise on parameter estimations is analyzed. Accurate formulas for estimating the standard deviations of these parameter estimations are also given, which are corroborated by means of computer simulations. Most of the derived formulas are more simples than the ones given in the scientific literature.

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1. Introduction

Parameter estimations of a multifrequency signal with high accuracy is fundamental in many engineering applications. Many methods are presented in the scientific literature about this task. These methods can be classified in time-domain methods and frequency-domain methods. Time-domain methods provide high frequency selectivity and high estimation accuracy, but require intensive computationally algorithms to determine the model order [1]. Frequency-domain methods use the Discrete Fourier Transform (DFT) to compute the spectrum of the multifrequency signal, which is exploited to estimate the multifrequency signal parameters [2–9]. These methods are characterized by its robustness toward signal model inaccuracies, but provide less frequency selectivity.

In a real case the coherent frequency relationships between all the frequencies contained in the input signal

and the sampling frequency are not meet, leading to the well-known leakage phenomena. The way used to reduce this problem is called 'windowing' and a frequency-domain method often used for estimating the multifrequency signal parameters under noncoherent sampling is the Interpolated DFT (IpDFT) method [4–9]. This method provides very accurate parameter estimates. The performances of the IpDFT method depend on the window used, which are cosine windows [6,7]. Among all these windows, the maximum sidelobe windows, known also as the class I Rife-Vincent windows [6], are frequently employed in the IpDFT method [10,11]. The IpDFT method with maximum sidelobe decay windows leads to very accurate estimates since in this case the parameters of a multifrequency signal can be estimated by analytical formulas [6–9]. The influence of white Gaussian noise and quantization noise on the parameter estimations has been analyzed in [7,8]. In these references the formulas for estimating the variances of the parameters of a multifrequency signal are given. It should be remarked that these formulas are complicated.

The goal of this paper is to derive accurate and more simple formulas for estimating the variances of the estima-

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tors of the parameters of a multifrequency signal obtained by IpDFT method with maximum sidelobe decay windows than the ones given in the scientific literature [7,8]. The organization of the paper is as follows: In Section 2 the theoretical expression of ones of the most important window parameters are derived for the H -term maximum sidelobe decay window. The analytical formulas for estimating the parameters of a multifrequency signal component are derived in Section 3. In Section 4, considering a multifrequency signal corrupted with additive white Gaussian noise, the formulas for estimating the standard deviations of the parameters are given. The accuracy of the proposed formulas is analyzed in Section 5 by means of computer simulations. Finally, Section 6 deals with the conclusions.

2. Characteristics of the maximum sidelobe decay windows

The maximum sidelobe decay windows belong to the cosine windows. A H -term maximum sidelobe decay window ($H \geq 2$) has the most rapidly decaying sidelobes, equal to $6(2H - 1)$ dB/octave, from all the H -term cosine windows defined by

$$w(m) = \sum_{h=0}^{H-1} (-1)^h a_h \cos\left(2\pi h \frac{m}{M}\right), \quad m = 0, 1, \dots, M-1, \quad (1)$$

where a_h are the window coefficients and M is the number of acquired samples.

If $w(m)$ is the maximum sidelobe decay window then the coefficients a_h have the following expressions [11]:

$$a_0 = \frac{C_{2H-2}^{H-1}}{2^{2H-2}}, \quad a_h = \frac{C_{2H-2}^{H-h-1}}{2^{2H-3}}, \quad h = 1, 2, \dots, H-1, \quad (2)$$

where $C_m^p = \frac{m!}{(m-p)!p!}$.

Ones of the most important parameters of a window are: *Normalized Peak Signal Gain* (NPSG), *Normalized Noise Power Gain* (NNPG) and *Equivalent Noise BandWidth* (ENBW).

- NPSG is defined by [12]

$$\text{NPSG} = \frac{\sum_{m=0}^{M-1} w(m)}{M}. \quad (3)$$

For a H -term cosine window, NPSG is equal to [13]

$$\text{NPSG} = a_0. \quad (4)$$

Using (2) it follows that for a H -term maximum sidelobe decay window, NPSG is equal to

$$\text{NPSG} = \frac{C_{2H-2}^{H-1}}{2^{2H-2}}. \quad (5)$$

- NNPG is defined by [12]

$$\text{NNPG} = \frac{\sum_{m=0}^{M-1} w^2(m)}{M}. \quad (6)$$

For a H -term cosine window, NNPG is equal to [13]

$$\text{NNPG} = a_0^2 + 0.5 \sum_{h=1}^{H-1} a_h^2. \quad (7)$$

Using (2) it follows that for a H -term maximum sidelobe decay window, NNPG is equal to

$$\text{NNPG} = \frac{1}{2^{4H-5}} \left[\sum_{h=0}^{H-1} \left(C_{2H-2}^h \right)^2 - 0.5 \left(C_{2H-2}^{H-1} \right)^2 \right]. \quad (8)$$

In order to simplify the NNPG expression, we use the following equality:

$$\sum_{h=0}^{2H-2} \left(C_{2H-2}^h \right)^2 = C_{4H-4}^{2H-2}, \quad (9)$$

which is true because the left side is the coefficient of the x^{2H-2} of the binomials product $(1+x)^{2H-2} \cdot (1+x)^{2H-2}$ and the right side is the coefficient of x^{2H-2} of the binomial $(1+x)^{4H-4}$. As the sum in (9) can be written as

$$\sum_{h=0}^{2H-2} \left(C_{2H-2}^h \right)^2 = 2 \sum_{h=0}^{H-2} \left(C_{2H-2}^h \right)^2 + \left(C_{2H-2}^{H-1} \right)^2, \quad (10)$$

using (9) and (10) it can be established that

$$\sum_{h=0}^{H-1} \left(C_{2H-2}^h \right)^2 = 0.5 \left[C_{4H-4}^{2H-2} + \left(C_{2H-2}^{H-1} \right)^2 \right]. \quad (11)$$

By replacing the above sum in (8) we obtain

$$\text{NNPG} = \frac{C_{4H-4}^{2H-2}}{2^{4H-4}}. \quad (12)$$

- ENBW is defined by [12]

$$\text{ENBW} = \frac{\text{NNPG}}{(\text{NPSG})^2}. \quad (13)$$

From (5) and (12) it follows that for a H -term maximum sidelobe decay window, ENBW is equal to

$$\text{ENBW} = \frac{C_{4H-4}^{2H-2}}{\left(C_{2H-2}^{H-1} \right)^2}. \quad (14)$$

As it can be seen from (5), (12) and (14) the derived expressions for NPSG, NNPG and ENBW depend only of H . Thus, these parameters can be very simple calculated using the derived expressions.

Another very important window parameter that reflects the reduction of the output signal-to-noise ratio (SNR) as a result of windowing and spectrum granularity is the *Scalloping Loss* (SL) defined by

$$\text{SL}(0.5) = \frac{|W(0.5)|}{W(0)}, \quad (15)$$

where $W(\lambda)$ is the Discrete-Time Fourier Transform (DTFT) of the window $w(m)$.

The DTFT of the H -term cosine window $w(m)$ is given by

$$W(\lambda) = \sin(\pi\lambda) e^{-j\pi\lambda} e^{j\frac{\pi}{M}\lambda} \sum_{h=0}^{H-1} (-1)^h 0.5 a_h \left[\frac{e^{-j\frac{\pi}{M}\lambda}}{\sin \frac{\pi}{M}(\lambda-h)} + \frac{e^{j\frac{\pi}{M}\lambda}}{\sin \frac{\pi}{M}(\lambda+h)} \right], \quad \lambda \in [0, M). \quad (16)$$

For $M \gg 1$, the DTFT of the H -term maximum sidelobe decay window can be approximated by [9]

$$W(\lambda) = \frac{M \sin(\pi\lambda)}{2^{2H-2} \pi \lambda} e^{-j\pi\lambda} e^{j\frac{\pi}{M}\lambda} \frac{(2H-2)!}{\prod_{h=1}^{H-1} (h^2 - \lambda^2)}. \quad (17)$$

Using (17), $SL(\delta)$ can be expressed by

$$SL(\delta) = \frac{\sin(\pi\delta)}{\pi\delta} \frac{[(H-1)!]^2}{\prod_{h=1}^{H-1} (h^2 - \delta^2)}. \quad (18)$$

From the above expression, it follows that $SL(\delta)$ depends only of δ and H .

After some calculus, $SL(0.5)$ can be written as

$$SL(0.5) = \frac{2^{4H-3} [(H-1)!]^4}{\pi(2H-1)!(2H-2)!}. \quad (19)$$

It should be pointed out that in the scientific literature [6–8] the parameters NNPG, ENBW and $SL(\delta)$ are calculated using their definition formulas, which is more complicated than by using the expressions (12), (14) and (18).

3. Analytical formulas of the parameter estimates

Now, let us consider a multifrequency signal $x(t)$ sampled at f_s frequency

$$x(m) = A_0 + \sum_{k=1}^K A_k \sin\left(2\pi \frac{f_k}{f_s} m + \varphi_k\right), \quad m = 0, 1, \dots, M-1, \quad (20)$$

where K is the number of frequency components, A_k , f_k and φ_k are, respectively, the amplitude, frequency and phase of the k th component, A_0 is the offset and M is the number of acquired samples.

As the sampling process was considered in noncoherent sampling mode, the spectrum of $x(m)$ is affected by the leakage errors. In order to reduce these errors, $x(m)$ is multiplied by a suitable window sequence $w(m)$. Thus, the DTFT of the resulting signal $x_w(m) = x(m) \cdot w(m)$ is given by

$$X_w(\lambda) = A_0 W(\lambda) + \sum_{k=1}^K \frac{A_k}{2j} [W(\lambda - \lambda_k) e^{j\varphi_k} - W(\lambda + \lambda_k) e^{-j\varphi_k}], \quad \lambda \in [0, M), \quad (21)$$

where λ represents the normalized frequency expressed in bin, $W(\lambda)$ is the DTFT of $w(m)$ and $\lambda_k = f_k/f_0$, in which $f_0 = f_s/M$.

If $W(\lambda)$ exhibits sidelobes with negligible level and if the minimum distance between spectral lines is more larger than the *Main Lobe Band Width* (MLBW) expressed in bin, then for $\lambda \cong \lambda_k$, (21) becomes

$$X_w(\lambda) \cong \frac{A_k}{2j} W(\lambda - \lambda_k) e^{j\varphi_k}, \quad k = 1, 2, \dots, K. \quad (22)$$

The relationship between the frequencies f_k and f_s is given by

$$\frac{f_k}{f_s} = \frac{\lambda_k}{M} = \frac{l_k + \delta_k}{M}, \quad k = 1, 2, \dots, K \quad (23)$$

in which l_k and δ_k are, respectively, the integer part and the fractional part of λ_k ($-0.5 \leq \delta_k < 0.5$).

Frequency estimation leads to estimate $\lambda_k = l_k + \delta_k$. As l_k is the index of the largest discrete spectrum module corre-

sponding to the k th component (easily determined by means of a maximum search routine), IpDFT method is used to estimate δ_k . For this purpose, the following rapport, α_k is evaluated

$$\alpha_k = \begin{cases} \frac{|X_w(l_k)|}{|X_w(l_k-1)|}, & \text{if } -0.5 \leq \delta_k < 0 \\ \frac{|X_w(l_k+1)|}{|X_w(l_k)|}, & \text{if } 0 \leq \delta_k < 0.5 \end{cases}. \quad (24)$$

It should be noted that if $|X_w(l_k-1)| > |X_w(l_k+1)|$ ($|X_w(l_k-1)| < |X_w(l_k+1)|$) then $-0.5 \leq \delta_k < 0$ ($0 < \delta_k < 0.5$). The reciprocity is also true, i.e. if $-0.5 \leq \delta_k < 0$ ($0 < \delta_k < 0.5$) then $|X_w(l_k-1)| > |X_w(l_k+1)|$ ($|X_w(l_k-1)| < |X_w(l_k+1)|$).

From (22) and (23) the rapport α_k becomes

$$\alpha_k \cong \begin{cases} \frac{|W(\delta_k)|}{|W(1+\delta_k)|}, & \text{if } -0.5 \leq \delta_k < 0 \\ \frac{|W(1-\delta_k)|}{|W(\delta_k)|}, & \text{if } 0 \leq \delta_k < 0.5 \end{cases}. \quad (25)$$

Using (17) after some calculus, it is obtained

$$\alpha_k \cong \begin{cases} \frac{H+\delta_k}{H-1-\delta_k}, & \text{if } -0.5 \leq \delta_k < 0 \\ \frac{H-1+\delta_k}{H-\delta_k}, & \text{if } 0 \leq \delta_k < 0.5 \end{cases}. \quad (26)$$

From the above expression it follows that δ_k can be estimated by

$$\hat{\delta}_k = \begin{cases} \frac{(H-1)\alpha_k - H}{\alpha_k + 1}, & \text{if } -0.5 \leq \delta_k < 0 \\ \frac{H\alpha_k - H + 1}{\alpha_k + 1}, & \text{if } 0 \leq \delta_k < 0.5 \end{cases}. \quad (27)$$

From (23) and (27) the frequency of the k th component can be estimated by

$$\hat{f}_k = \begin{cases} \left(l_k + \frac{(H-1)\alpha_k - H}{\alpha_k + 1}\right) \cdot f_0, & \text{if } -0.5 \leq \delta_k < 0 \\ \left(l_k + \frac{H\alpha_k - H + 1}{\alpha_k + 1}\right) \cdot f_0, & \text{if } 0 \leq \delta_k < 0.5 \end{cases}. \quad (28)$$

Using (17) and (22) the amplitude of the k th component can be estimated by

$$\hat{A}_k = \frac{2^{2H-1} \pi \delta_k |X_w(l_k)|}{M \sin(\pi \delta_k) (2H-2)!} \prod_{h=1}^{H-1} (h^2 - \delta_k^2). \quad (29)$$

The formula (41) (see next chapter) is used in [6,8] to estimate A_k . This formula gives the same estimate for A_k as (28), but is more complicated since involves the calculation of $D(\delta_k)$ [see (43)].

From an accurate estimation of the phase associated with the k th component, (22) is used in which the window's DTFT is computed by (16)

$$\hat{\varphi}_k = \text{Phase}\{X_w(l_k)\} - \pi \delta_k + \pi \frac{\delta_k}{M} - \frac{\pi}{2} \text{sign}(\delta_k) - \text{Phase}\{W_0(-\delta_k)\}, \quad (30)$$

where

$$W_0(\lambda) = \sum_{h=0}^{H-1} (-1)^h 0.5 a_h \left[\frac{e^{-j\frac{\pi}{M}h}}{\sin \frac{\pi}{M}(\lambda - h)} + \frac{e^{j\frac{\pi}{M}h}}{\sin \frac{\pi}{M}(\lambda + h)} \right]$$

sign() is the sign function,

$$\text{sign}(\delta_k) = \begin{cases} -1, & \text{if } -0.5 \leq \delta_k < 0 \\ 1, & \text{if } 0 < \delta_k < 0.5 \end{cases}.$$

4. The influence of a white Gaussian noise on the parameter estimation

Due to its real-life importance a white Gaussian noise of variances σ_n has been added to $x(m)$. Nevertheless, this implies a statistical behaviour for all the estimators presented in the previous section. The aim of this section is to study this behaviour by means of standard deviation analysis.

4.1. Standard deviation analysis of the frequency estimator

The variances of the spectral lines (except the one corresponding to the DC component) are equal to [7]

$$\sigma_{X_w}^2 = \frac{M}{2} \text{NNPG} \cdot \sigma_n^2. \quad (31)$$

The correlation coefficient between two neighboring spectral lines $|X_w(i)|$ and $|X_w(i+1)|$ ($i \neq 0$) is given by [14]

$$r = \rho(|X_w(i)|, |X_w(i+1)|) = \frac{|a_0 a_1 + 0.5 \sum_{h=1}^{H-2} a_h a_{h+1}|}{\text{NNPG}}. \quad (32)$$

From the relationships (24), (27), (31) and (32) the variance of the δ_k estimation is expressed by

$$\sigma_{\delta_k}^2 \cong \begin{cases} \frac{(2H-1)^2}{(1+\alpha_k)^4} \cdot \frac{(1-2\alpha_k r + \alpha_k^2)}{(|X_w(l_k-1)|)^2} \cdot \sigma_{X_w}^2, & \text{if } -0.5 \leq \delta_k < 0 \\ \frac{(2H-1)^2}{(1+\alpha_k)^4} \cdot \frac{(1-2\alpha_k r + \alpha_k^2)}{(|X_w(l_k)|)^2} \cdot \sigma_{X_w}^2, & \text{if } 0 \leq \delta_k < 0.5 \end{cases} \quad (33)$$

where α_k is calculated by (26) and $r = \frac{|a_0 a_1 + 0.5 \sum_{h=1}^{H-2} a_h a_{h+1}|}{\text{NNPG}}$.

From (2) the nominator of r becomes

$$\left| a_0 a_1 + 0.5 \sum_{h=1}^{H-2} a_h a_{h+1} \right| = \frac{1}{2^{4H-5}} \sum_{h=0}^{H-2} C_{2H-2}^h C_{2H-2}^{h+1}. \quad (34)$$

To simplify r expression, let us consider the following equality [15]

$$\sum_{h=0}^{2H-3} C_{2H-2}^h C_{2H-2}^{h+1} = C_{4H-4}^{2H-3}, \quad (35)$$

where the sum can be written as follow:

$$\sum_{h=0}^{2H-3} C_{2H-2}^h C_{2H-2}^{h+1} = 2 \sum_{h=0}^{H-2} C_{2H-2}^h C_{2H-2}^{h+1}. \quad (36)$$

Thus, using (35) and (36) the nominator of r becomes

$$\left| a_0 a_1 + 0.5 \sum_{h=1}^{H-2} a_h a_{h+1} \right| = \frac{1}{2^{4H-4}} C_{4H-4}^{2H-3}. \quad (37)$$

Finally, using (12) and (36) r is equal to

$$r = \frac{2H-2}{2H-1}. \quad (38)$$

Using (12), (17) and (22) the standard deviation of δ_k estimation can be approximated by

$$\sigma_{\delta_k} = \begin{cases} \frac{2(H-1-\delta_k)(H+\delta_k)}{(2H-1)!} \frac{\pi \delta_k}{\sin(\pi \delta_k)} \frac{\prod_{h=1}^{H-1} (h^2 - \delta_k^2)}{A_k} \\ \sqrt{\frac{C_{4H-4}^{2H-2}}{2M}} (1 - 2\alpha_k r + \alpha_k^2) \cdot \sigma_n, & \text{if } -0.5 \leq \delta_k < 0 \\ \frac{2(H-\delta_k)^2}{(2H-1)!} \frac{\pi \delta_k}{\sin(\pi \delta_k)} \frac{\prod_{h=1}^{H-1} (h^2 - \delta_k^2)}{A_k} \\ \sqrt{\frac{C_{4H-4}^{2H-2}}{2M}} (1 - 2\alpha_k r + \alpha_k^2) \cdot \sigma_n, & \text{if } 0 \leq \delta_k < 0.5 \end{cases} \quad (39)$$

Using (26), after some calculus, it can be established that

$$\sigma_{\delta_k} \cong \frac{2(H-|\delta_k|)}{(2H-1)!} \frac{\pi \delta_k}{\sin(\pi \delta_k)} \frac{\prod_{h=1}^{H-1} (h^2 - \delta_k^2)}{A_k} \sqrt{\frac{C_{4H-4}^{2H-2}}{2M}} \times \sqrt{\frac{2(4H-3)(\delta_k^2 - |\delta_k|) + 2H^2 - 1}{2H-1}} \cdot \sigma_n. \quad (40)$$

The above formula is more simple than the one given in [7, relationship (23)], which requires the calculation of DTFT of $w^2(m)$ and of $\frac{\partial \delta_k}{\partial \alpha_k}$.

Moreover, the same relationships have been also, obtained when the signal $x(m)$ is corrupted with quantization noise.

4.2. Standard deviation analysis of the amplitude estimator

Concerning the standard deviation analysis of the amplitude estimator, $\hat{A}_k$ given by [6,8] is used

$$\hat{A}_k = \frac{2\pi}{N} \frac{|X_w(l_k)|}{D(\delta_k)} \quad (41)$$

or

$$\hat{A}_k = \frac{2\pi}{N} \frac{|X_w(l_k-1)|}{D(1+\delta_k)}, \quad (42)$$

where

$$D(\delta_k) = \delta_k \sin(\pi \delta_k) \sum_{h=0}^{H-1} (-1)^h \frac{a_h}{\delta_k^2 - h^2}. \quad (43)$$

It is interesting to note that the sum of this relation can be expressed as [9]

$$\sum_{h=0}^{H-1} (-1)^h \frac{a_h}{\delta_k^2 - h^2} = \frac{(-1)^{H-1} (2H-2)!}{2^{2H-2} \prod_{h=0}^{H-1} (\delta_k^2 - h^2)}. \quad (44)$$

First, the case $0 \leq \delta_k < 0.5$ was considered and using (41) the variance of $\hat{A}_k$ can be expressed by

$$\sigma_{\hat{A}_k}^2 = \left[\frac{\hat{A}_k^2}{(|X_w(l_k)|)^2} \sigma_{X_w}^2 + \frac{\hat{A}_k^2}{D^2} \left(\frac{\partial D}{\partial \delta_k} \right)^2 \sigma_{\delta_k}^2 - \frac{2\hat{A}_k}{|X_w(l_k)| D} \frac{\partial D}{\partial \delta_k} \sigma_{X_w} \sigma_{\delta_k} \rho(|X_w(l_k)|, \delta_k) \right], \quad (45)$$

where $D = D(\delta_k)$ and $\rho(|X_w(l_k)|, \delta_k)$ is the correlation coefficient between $|X_w(l_k)|$ and δ_k . Note that $\frac{\partial D}{\partial \delta_k}$ is expressed by

$$\frac{\partial D}{\partial \delta_k} = (\sin(\pi\delta_k) + \pi\delta_k \cos(\pi\delta_k)) \sum_{h=0}^{H-1} (-1)^h \frac{a_h}{\delta_k^2 - h^2} - 2\delta_k^2 \times \sin(\pi\delta_k) \sum_{h=0}^{H-1} (-1)^h \frac{a_h}{(\delta_k^2 - h^2)^2}.$$

From (24) and (26) it follows that

$$|X_w(l_k + 1)| = \alpha_k |X_w(l_k)| \cong \frac{H-1+\delta_k}{H-\delta_k} |X_w(l_k)|. \quad (46)$$

From the above relationship it can be established

$$\sigma_{X_w}^2 \cong \alpha_k^2 \sigma_{X_w}^2 + (1 - 2\alpha_k r + \alpha_k^2) \sigma_{X_w}^2 + 2\alpha_k \times \sqrt{1 - 2\alpha_k r + \alpha_k^2} \sigma_{X_w}^2 \rho(|X_w(l_k)|, \delta_k). \quad (47)$$

Thus, it follows:

$$\rho(|X_w(l_k)|, \delta_k) \cong \frac{r - \alpha_k}{\sqrt{1 - 2\alpha_k r + \alpha_k^2}}. \quad (48)$$

Based on the relationships (12), (29), (31), (40) and (48) after some calculus we obtain

$$\sigma_{\hat{A}_k} \cong \frac{2\pi\delta_k \prod_{h=1}^{H-1} (h^2 - \delta_k^2)}{(2H-2)! \sin(\pi\delta_k)} \sqrt{\frac{C_{4H-4}^{2H-2}}{2M}} \sqrt{1 + \frac{(2H-1)^2}{(1+\alpha_k)^4} \frac{1}{D^2} \left(\frac{\partial D}{\partial \delta_k}\right)^2} (1 - 2\alpha_k r + \alpha_k^2) - \frac{2(2H-1)(r - \alpha_k)}{(1+\alpha_k)^2} \frac{1}{D} \frac{\partial D}{\partial \delta_k} \cdot \sigma_n. \quad (49)$$

For $-0.5 \leq \delta_k < 0$ a similar procedure is used to calculate $\sigma_{\hat{A}_k}$ with the relationship (42). All the relationships derived concerning $\sigma_{\hat{A}_k}$ for $0 \leq \delta_k < 0.5$ remains valid for $-0.5 \leq \delta_k < 0$, but with δ_k replaced by $1 + \delta_k$.

Thus, standard deviation of A_k estimation can be approximated by

$$\sigma_{\hat{A}_k} \cong \frac{2\pi\tilde{\delta}_k \prod_{h=1}^{H-1} (h^2 - \tilde{\delta}_k^2)}{(2H-2)! \sin(\pi\tilde{\delta}_k)} \sqrt{\frac{C_{4H-4}^{2H-2}}{2M}} \sqrt{1 + \frac{(2H-1)^2}{(1+\alpha_k)^4} \frac{1}{D^2} \left(\frac{\partial D}{\partial \delta_k}\right)^2} (1 - 2\alpha_k r + \alpha_k^2) - \frac{2(2H-1)(r - \alpha_k)}{(1+\alpha_k)^2} \frac{1}{D} \frac{\partial D}{\partial \delta_k} \cdot \sigma_n, \quad (50)$$

where $D = D(\tilde{\delta}_k) = \tilde{\delta}_k \sin(\pi\tilde{\delta}_k) \sum_{h=0}^{H-1} (-1)^h \frac{a_h}{\tilde{\delta}_k^2 - h^2}$,

with $\tilde{\delta}_k = \begin{cases} 1 + \delta_k, & \text{if } -0.5 \leq \delta_k < 0 \\ \delta_k, & \text{if } 0 \leq \delta_k < 0.5. \end{cases}$

The derived formula for $\sigma_{\hat{A}_k}$ is complicated, but not as the formula given in [7relationship (24)], which requires the calculation of DTFT of $w^2(m)$, of the derivate of $W(\delta_k)$ and of $\frac{\partial \delta_k}{\partial \delta_k}$.

It should be noted that the same relationships have been also, obtained when the signal $x(m)$ is corrupted with quantization noise.

In [8] the definition formula (45) is given to calculate $\sigma_{\hat{A}_k}$, which is evident more complicated than (50). Moreover, from (45) we cannot conclude that $\sigma_{\hat{A}_k}$ not depends of A_k , as we can conclude from (50).

4.3. Standard deviation analysis of the phase estimator

For large M the standard deviation of φ_k estimation can be approximated by [7]

$$\sigma_{\varphi_k} \cong \sqrt{\frac{2}{M} \frac{\sigma_q^2}{A_k^2} \frac{\text{ENBW}}{[\text{SL}(\delta_k)]^2} + \pi^2 \sigma_{\delta_k}^2}, \quad (51)$$

where $\text{SL}(\delta_k)$ is computed by (18).

It should be noted that the above relationship has been also, obtained when the signal $x(m)$ is corrupted with quantization noise.

5. Computer simulations

The aim of this section is to verify by means of computer simulations the accuracy of the analytical formulas derived in Section 3 dedicated to the parameter estimations of a multifrequency component. The accuracy of the formulas derived in Section 4 for estimating the standard deviations of the estimators of the parameters of a multifrequency signal is also examined. For this purpose a single component signal with $A = 1$, $\varphi = 0.5$ rad and $l = 73$ is used in simulation. The number of acquired samples is equal to $M = 1024$. An additive white Gaussian noise characterized by zero mean and standard deviation σ_n is added to the

signal. σ_n is established in order to obtain a signal-to-noise ratio (SNR) equal to 45 dB. δ varies in the range $[-0.5, 0.5]$ with an increment of 0.04. For each δ value 10,000 runs are used.

Fig. 1 shows the modulus of the bias of signal parameters as a function of δ . The signal parameters are estimated by the analytical formulas derived in Section 3. The three-

term maximum sidelobe decay window is employed in the IpDFT method.

Simulations show that when the signal parameters are estimated by the analytical formulas (27), (29) and (30) negligible bias is obtained.

Fig. 2 presents the standard deviations of signal parameter estimates calculated by the formulas derived in Section 4 and the ones obtained by simulation as a function of δ . The three-term maximum sidelobe decay window is employed in the IpDFT method.

The derived standard deviations related to the δ , A and φ differs from the simulated ones by maximum 1.43%, 1.82% and 1.68%, respectively. Thus, the standard deviations related to the δ , A and φ can be accurately estimated by the formulas (40), (50) and (51), respectively.

Simulations have been carried out also for different values of M , σ_n (SNR higher than 20 dB) and H (equal to 2 and 4) and similar results with the ones above presented have been obtained.

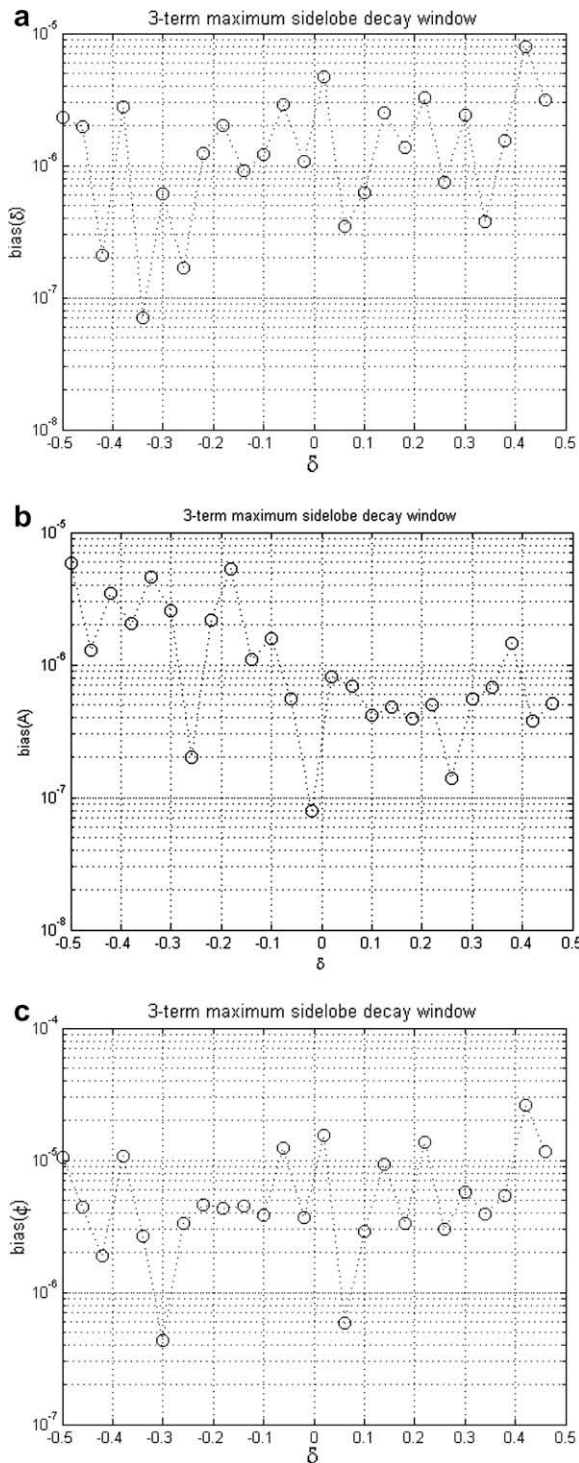


Fig. 1. The modulus of the bias of signal parameters as a function of δ when the three-term maximum sidelobe decay window is used in IpDFT method.

6. Conclusions

In this paper, a study concerning the estimation of the parameters of a multifrequency signal by IpDFT method

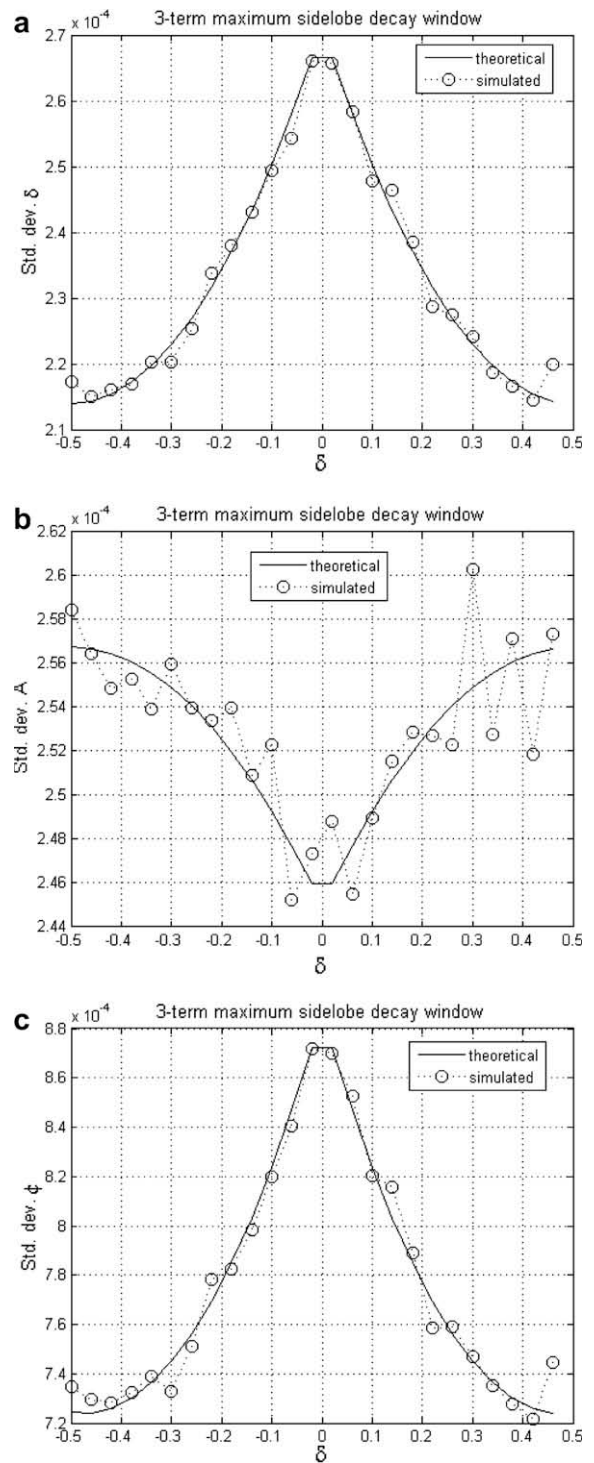


Fig. 2. The derived standard deviations of the signal parameter estimates and the ones obtained by simulation as a function of δ when the three-term maximum sidelobe decay window is used in IpDFT method.

with maximum sidelobe decay windows is made. The theoretical expressions of ones of the most important parameters of a window – NPSG, NNPG and ENBW are derived for

a H -term maximum sidelobe decay window. These expressions depend only of H . Moreover, has been derived an expression for the window's parameter $SL(\delta)$, which depends only of δ and H . Thus, by the expressions derived the above mentioned window parameters can be very easily calculated. The analytical formulas for estimating the parameters of a multifrequency signal component by IpDFT method with the H -term maximum sidelobe decay window are given. The analytical formula for amplitude estimate is more simples than the ones presented in scientific literature [6,8]. Furthermore, for a multifrequency signal corrupted with white Gaussian noise the formulas for the standard deviations of the parameter estimates are given. These formulas remain also, valid when the multifrequency signal is corrupted with quantization noise. The formulas for the standard deviations of δ_k and A_k estimates are more simples than the ones presented in the scientific literature [7,8].

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